
roboRIO 2.0

Embedded

Controller for

FRC

Specifications

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roboRIO 2.0 Embedded Controller for FRC Specifications

roboRIO 2.0 Specifications

These specifications apply to the roboRIO 2.0 Embedded Controller for FRC.

These specifications are typical for the 0° C to 40° C operating temperature range unless otherwise noted.

Looking For Something Else?

For information not found in the specifications for your product, such as operating instructions, browse *Related Information*.

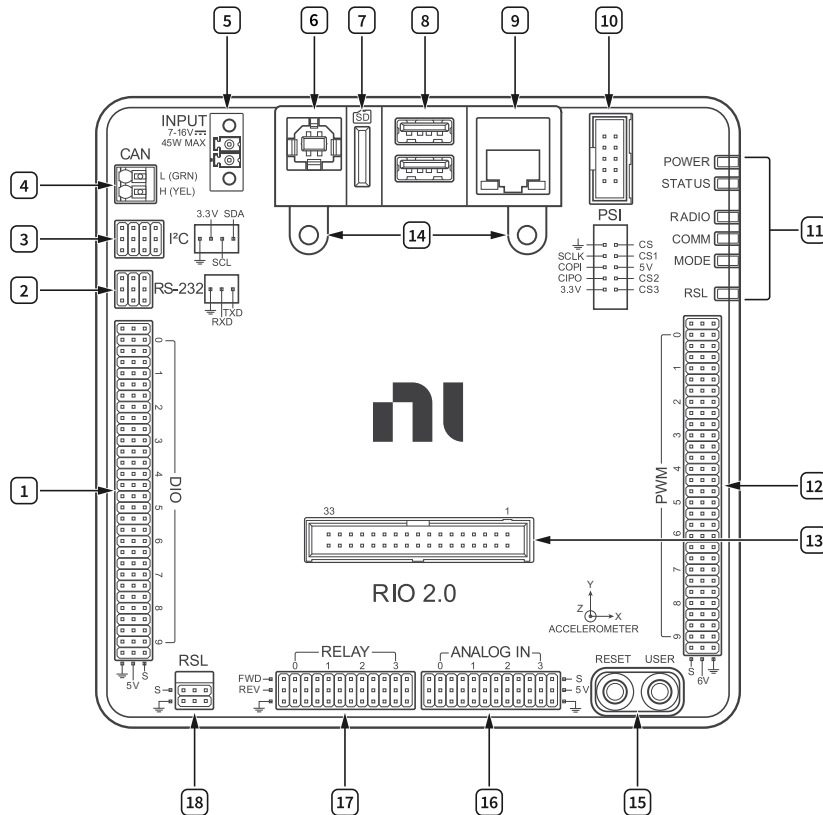
Related information:

- [roboRIO 2.0 Embedded Controller for FRC User Manual](#)
- [roboRIO Product Certifications](#)
- [Discussion Forums](#)
- [NI Learning Center](#)

roboRIO 2.0 Pinouts

Use the pinouts to connect to terminals on the roboRIO 2.0.

Figure 1. roboRIO 2.0 I/O



1. Digital input and output (DIO) port	2. RS-232 port	3. I2C port
4. CAN port	5. Power connector	6. USB Device port
7. microSD card slot	8. USB Host ports	9. Ethernet port
10. Serial peripheral interface bus (SPI) port	11. LEDs	12. Pulse-width modulation (PWM) port
13. myRIO Expansion Port (MXP)	14. MXP retention mount	15. User and Reset buttons
16. Analog input (AI) port	17. Relay port	18. Robot signal light (RSL) port

MXP Pinout

The following figure and table describe the MXP pins and signals. Note that some pins carry both primary and secondary functions.

Figure 2. MXP Pinout

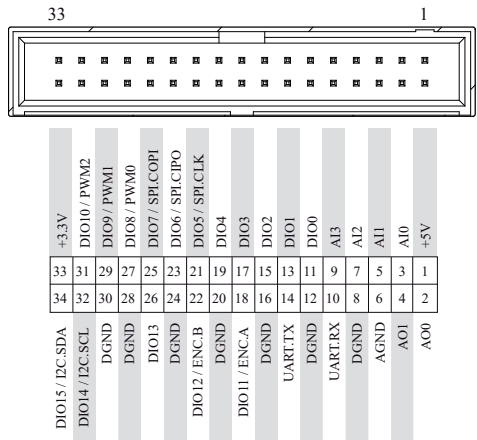


Table 1. MXP Signal Descriptions

Signal Name	Reference	Direction	Description
+5V	DGND	Output	+5 V power output.
AI <0..3>	AGND	Input	0 V to 5 V, referenced, single-ended AI channels.
AO <0..1>	AGND	Output	0 V to 5 V referenced, single-ended AO.
AGND	—	—	Reference for AI and AO.
+3.3V	DGND	Output	+3.3 V power output.
DIO <0..15>	DGND	Input or Output	General-purpose digital lines with 3.3 V output, 3.3 V-/5 V-compatible input.
UART.RX	DGND	Input	UART receive input. UART lines are electrically identical to DIO lines.
UART.TX	DGND	Output	UART transmit output. UART lines are electrically identical to DIO lines.
DGND	—	—	Reference for digital signals, +5 V, and +3.3

Signal Name	Reference	Direction	Description
			V.

CAN Port Pinout

The following figure and table describe the CAN port pins and signals.

Figure 3. CAN Port Pinout

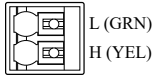


Table 2. Signal Descriptions

Signal Name	Direction	Description
L (GRN)	Input/Output	CAN bus differential low signal.
H (YEL)	Input/Output	CAN bus differential high signal.



Note The roboRIO 2.0 contains an internal 120 Ω termination resistor between L (GRN) and H (YEL).

I²C Port Pinout

The following figure and table describe the I²C port pins and signals.

Figure 4. I²C Port Pinout

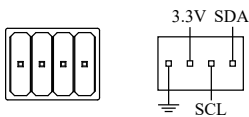


Table 3. Signal Descriptions

Signal Name	Direction	Description
GND	—	Reference for digital lines and +3.3 V power output.
3.3V	Output	+3.3 V power output.
SCL	Input or Output	I ² C lines with 3.3 V output, 3.3

Signal Name	Direction	Description
SDA	Input or Output	V-/5 V-compatible input.

RS-232 Port Pinout

The following figure and table describe the RS-232 port pins and signals.

Figure 5. RS-232 Serial Port Pinout

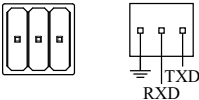


Table 4. Signal Descriptions

Signal Name	Direction	Description
TXD	Output	Serial transmit output with ± 5 V to ± 15 V signal levels.
RXD	Input	Serial receive input with ± 15 V input voltage range.
GND	—	Reference for digital lines.

DIO Port Pinout

The following figure and table describe the DIO port pins and signals.

Figure 6. DIO Port Pinout

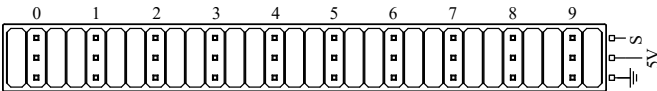


Table 5. Signal Descriptions

Signal Name	Direction	Description
S (DIO) <0..9>	Input/Output	General-purpose digital lines with 3.3 V output, 3.3 V-/5 V-compatible input.
5V	Output	+5 V power output.
GND	—	Reference for digital lines and +5 V power output.

RSL Port Pinout

The following figure and table describe the RSL port pins and signals.

Figure 7. RSL Port Pinout

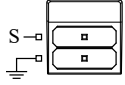


Table 6. Signal Descriptions

Signal Name	Direction	Description
S	Output	Switched power output to drive RSL when RSL is enabled. The voltage level depends on the connected input voltage. RSL current is limited at 120 mA.
GND	—	Reference for S.

Relay Port Pinout

The following figure and table describe the Relay port pins and signals.

Figure 8. Relay Port Pinout

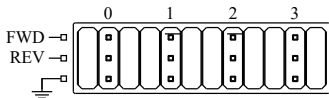


Table 7. Signal Descriptions

Signal Name	Direction	Description
FWD <0..3>	Output	Relay digital lines with 5 V output.
REV <0..3>	Output	Relay digital lines with 5 V output.
GND	—	Reference for digital lines.

AI Port Pinout

The following figure and table describe the AI port pins and signals.

Figure 9. AI Port Pinout

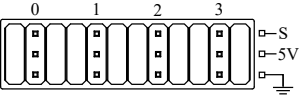


Table 8. Signal Descriptions

Signal Name	Direction	Description
S (AI) <0..3>	Input	0 V to 5 V referenced, single-ended AI channels.
5V	Output	+5 V power output.
GND	—	Reference for AI and +5 V power.

PWM Port Pinout

The following figure and table describe the PWM port pins and signals.

Figure 10. PWM Port Pinout

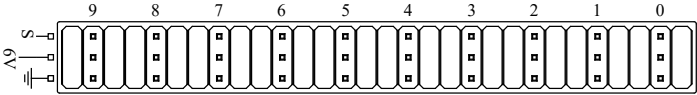


Table 9. Signal Descriptions

Signal Name	Direction	Description
S (PWM) <0..9>	Output	PWM digital lines with 5 V output.
6V	Output	+6 V power output for servos only.
GND	—	Reference for digital lines and +6 V power output.

SPI Port Pinout

The following figure and table describe the SPI port pins and signals.

Figure 11. SPI Port Pinout

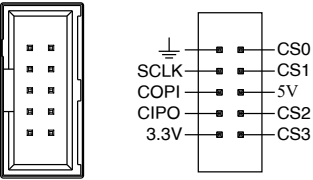


Table 10. Signal Descriptions

Signal Name	Direction	Description
3.3V	Output	+3.3 V power output.
5V	Output	+5 V power output.
CS <0..3>	Output	SPI with 3.3 V output, 3.3 V-/5 V-compatible input.
SCLK	Output	
COPI	Output	
CIPO	Input	
GND	—	Reference for digital lines and +3.3 V and +5.5 V power output.

Processor

Type	Xilinx Z-7020 All Programmable SoC dual-core ARM Cortex-A9
Speed	866 MHz
Cores	2

Memory

Nonvolatile	16 MB
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microSD	4 GB
DDR3	
Amount	512 MB
Clock frequency	533 MHz
Data bus width	16 bits

For information about the life span of the nonvolatile memory and about best practices for using nonvolatile memory, visit ni.com/info and enter the Info Code SSDBP.

FPGA

Type	Xilinx Z-7020
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Network

Network interface	10BaseT and 100BaseTX Ethernet
Compatibility	IEEE 802.3
Communication rates	10 Mbps, 100 Mbps, auto-negotiated
Maximum cabling distance	100 m/segment

USB Ports

Host	
Number of ports	2
Type	USB 2.0 Hi-Speed
VBus current	900 mA maximum per port
Device	
Number of ports	1
Type	USB 2.0 Hi-Speed

Analog Input

Aggregate sample rate	500 kS/s
Resolution	12 bits
Overvoltage protection	± 16 V
Expansion port configuration	4 single-ended channels
Integrated AI connector configuration	4 single-ended channels

Input impedance	>500 k Ω acquiring at 500 kS/s, 1 M Ω powered on and idle, 4.7 k Ω powered off
Recommended source impedance	3 k Ω or less
Nominal range	0 V to +5 V
Absolute accuracy	± 50 mV
Bandwidth	20 kHz minimum, >50 kHz typical

Analog Output

Aggregate maximum update rate	345 kS/s
Resolution	12 bits
Overload protection	± 16 V
Startup voltage	0 V after FPGA initialization
Configuration	2 single-ended channels on expansion port
Range	0 V to +5 V

Absolute accuracy	50 mV
Current drive	3 mA
Slew rate	0.3 V/ μ s

Digital I/O

Number of lines	
Expansion port	16 DIO lines; one UART
Integrated DIO, I²C, and SPI bus	
DIO lines	10 DIO lines
I ² C lines	1 SDA and 1 CLK
SPI lines	Drives up to four devices
Direction control	Each DIO line individually programmable as input or output
Logic level	5 V compatible LVTTTL input; 3.3 V LTTL output

Input logic levels	
Input low voltage, V_{IL}	0.0 V min; 0.8 V max

Input high voltage, V_{IH}	2.0 V min; 5.25 V max
Output logic levels	
Output low voltage, V_{OL} , sinking 4 mA	0.0 V min; 0.4 V max
Output high voltage, V_{OH} , sourcing 4 mA	2.4 V min; 3.465 V max

Minimum pulse width	20 ns
Maximum frequencies for secondary digital functions	
SPI	4 MHz
I ² C	400 kHz
UART lines	
Maximum baud rate	230,400 bps
Data bits	5, 6, 7, 8
Stop bits	1, 2
Parity	Odd, Even, Mark, Space
Flow control	XON/XOFF

RS-232 Serial Port

Maximum baud rate	115,200 bps
Data bits	5, 6, 7, 8
Stop bits	1, 2
Parity	Odd, Even, Mark, Space
Flow control	XON/XOFF
Logic level	
Standard	Meets or exceeds TIA/EIA-232-F voltage levels
Receiver input voltage	+30 V maximum
Driver output high voltage	5 V minimum
Driver output low voltage	-5 V maximum

PWM and Relay Lines

PWM port	10 PWM lines
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Relay port	4 forward; 4 reverse
Direction control	Output only
Logic level	5 V output
Maximum output current	
PWM	15.0 mA
Relay	7.5 mA
Series resistor in each output path	
PWM	330 Ω
Relay	680 Ω
Output high voltage, V_{OH}	
PWM sourcing 0.1 mA	4.75 V minimum; 5.25 V maximum
Relay sourcing 0.1 mA	4.75 V minimum; 5.25 V maximum
Output low voltage, V_{OL}	
PWM sinking 0.1 mA	0.0 V minimum; 0.25 V maximum
Relay sourcing 0.1 mA	0.0 V minimum; 0.25 V maximum

Maximum frequency	150 kHz
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RSL

RSL port	Switched VIN output
Voltage range	7 V to 16 V (VIN)
Current range	120 mA maximum

Accelerometer

Number of axes	3
Range	± 8 g
Resolution	12 bits
Sample rate	800 S/s
Noise	3.9 mg _{ms} typical at 25° C

Power Output

+6.0 V power output

Output voltage	5.5 V to 6.1 V
Output voltage with load >360 mA	5.75 V to 6.1 V
Maximum current	2.2 A total
+5.0 V power output	
Output voltage with and without load	4.7 V to 5.25 V
Maximum current	1.0 A total
+3.3 V power output	
Output voltage with and without load	3.1 V to 3.465 V
Maximum current	1.225 A total

Brownout Threshold

Default threshold	7 V
User-configurable threshold	4.5 V to 12 V
Tolerance	± 0.35 V
Hysteresis	0.85 V

Power Requirements

The roboRIO 2.0 requires a power supply connected to the power connector.

Power supply voltage range	7 VDC to 16 VDC
Maximum power consumption	45 W
Typical idle power consumption	5 W

Physical Characteristics

Dimensions	14.6 cm x 14.3 cm x 3.5 cm (5.75 inch x 5.64 inch x 1.37 inch)
Weight	335 g (11.82 oz)

Environmental Characteristics

Temperature	
Operating	0 °C to 40 °C
Storage	-20° C to 70° C
Humidity	
Operating	10% RH to 90% RH, noncondensing

Storage	10% RH to 90% RH, noncondensing	
Pollution Degree		2
Maximum altitude		2,000 m
Shock and Vibration		
Operating vibration		
Random	5 g _{RMS} , 10 Hz to 500 Hz	
Sinusoidal	5 g, 10 Hz to 500 Hz	
Operating shock	30 g, 11 ms half-sine; 50 g, 3 ms half-sine; 18 shocks at 6 orientations	

Environmental Standards

This product meets the requirements of the following environmental standards for electrical equipment.

- IEC 60068-2-1 Cold
- IEC 60068-2-2 Dry heat
- IEC 60068-2-78 Damp heat (steady state)
- IEC 60068-2-64 Random operating vibration
- IEC 60068-2-6 Sinusoidal operating vibration
- IEC 60068-2-27 Operating shock



Note To verify marine approval certification for a product, refer to the product label or visit ni.com/certification and search for the certificate.

Safety Compliance Standards

This product is designed to meet the requirements of the following electrical equipment safety standards for measurement, control, and laboratory use:

- IEC 61010-1, EN 61010-1
- UL 61010-1, CSA C22.2 No. 61010-1



Note For safety certifications, refer to the product label or the [Product Certifications and Declarations](#) section.

EMC Standards

This product meets the requirements of the following EMC standards for electrical equipment for measurement, control, and laboratory use:

- EN 61326-1 (IEC 61326-1): Class A emissions; Basic immunity
- EN 55011 (CISPR 11): Group 1, Class A emissions
- AS/NZS CISPR 11: Group 1, Class A emissions



Note Group 1 equipment is any industrial, scientific, or medical equipment that does not intentionally generate radio frequency energy for the treatment of material or inspection/analysis purposes.



Note In Europe, Australia, and New Zealand (per CISPR 11) Class A equipment is intended for use in non-residential locations.

Environmental Management

NI is committed to designing and manufacturing products in an environmentally responsible manner. NI recognizes that eliminating certain hazardous substances from our products is beneficial to the environment and to NI customers.

For additional environmental information, refer to the *Engineering a Healthy Planet* web page at ni.com/environment. This page contains the environmental regulations

and directives with which NI complies, as well as other environmental information not included in this document.

Product Certifications and Declarations

Refer to the product Declaration of Conformity (DoC) for additional regulatory compliance information. To obtain product certifications and the DoC for NI products, visit ni.com/product-certifications, search by model number, and click the appropriate link.